Swiss Reality Check

An open benchmark of Swiss knowledge, cultural framing, and Alemannic dialect — plus a measurable retrieval fix for Apertus 1.5. Hack Apertus · Track 1B Swiss Voices.

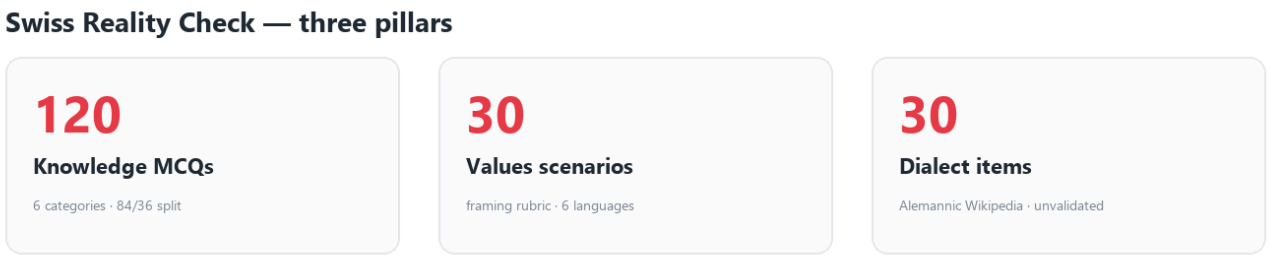


Figure A. Benchmark scale. Knowledge answers are verified against official Swiss sources; distractors are real neighbouring-country answers, not inventions.

1. The problem

Open multilingual models are trained overwhelmingly on global and English material. For Switzerland — four national languages, a distinct legal system, Alemannic dialects that are not Standard German, and direct-democracy instruments with no EU equivalent — that produces a specific, testable failure: **fluent, confident answers that belong somewhere else**. The question is not whether models *can* know Swiss facts. It is whether you can tell, on evidence, when they do not.

2. The measurement

- **Knowledge (120 MCQs)** — politics, law, health, geography, services, history (20 each). Stratified 84/36 split, seed 42. Test set sealed (SHA-256 df819e8c…).
- **Values (30 scenarios + multilingual pairs)** — rubric scores framing and balance, not political position. 73 judged records on the test run.
- **Dialect (30 items)** — verbatim Alemannic Wikipedia sentences; all items still carry `validated_by: null` pending native-speaker sign-off.

Headline (held-out test, Apertus 1.5 8B Q4_K_M): 55.6% → 86.1% with BM25 RAG (+30.6 pp, McNemar p=0.001). Apertus 70B was not evaluated (8 GB VRAM scope drop).

3. Method

3.1 Model & serving

Apertus 1.5 text extract, Q4_K_M GGUF, served via llama.cpp HTTP (`temperature=0.0`, seed 42). Backend stamped `is_mock: false` on every artefact. The original design intended an 8B vs 70B comparison; 70B was dropped because the evaluation GPU has 8 GB VRAM and cannot host it. No 70B accuracy is reported anywhere in this PDF.

3.2 Improvement: BM25 RAG (not a fact-stuffed prompt)

Corpus: 100 harvested official / Alemannic source pages → 189 non-dialect chunks for knowledge retrieval. Retriever: pure-Python BM25, `top_k=3`. A plan-proposed system prompt listing Swiss facts would have answered **25/84 (30%)** of the development set from the prompt alone. The shipped behavioural prompt answers **0/84** — enforced by a hard contamination gate before every RAG run.

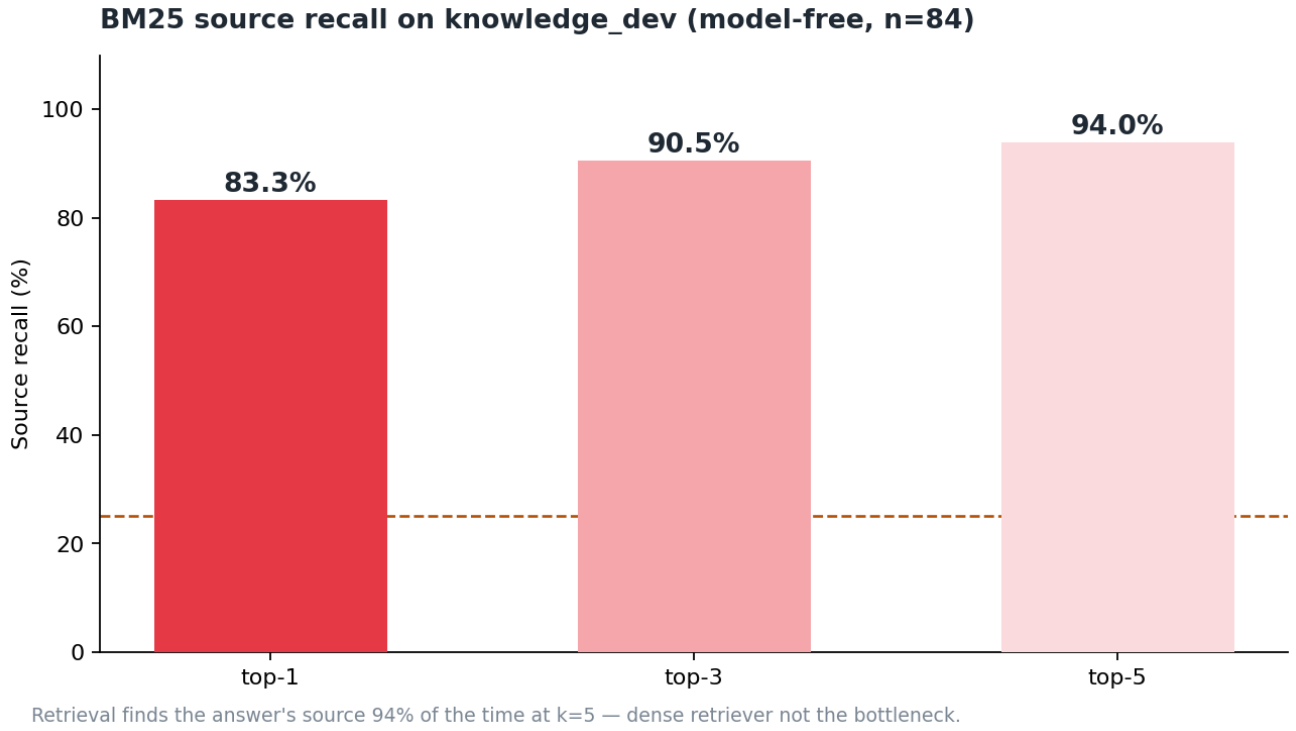


Figure B. Model-free source recall on knowledge_dev. Retrieval is not the bottleneck (94.0% @5); a dense embedder was not worth installing before a real baseline existed.

3.3 Evaluation integrity

- Same scorer functions for dev and sealed test (no reimplemented parsers).
- Append-only seal access log; sealed file hash verified.
- Disclosure: the sealed set was opened more than once due to a since-fixed `--resume` bug. Reported numbers are from the completed arms (baseline 55.6%, improved 86.1%), not a mean over opens. See eval/README.md.

4. Main result — knowledge (held-out test)

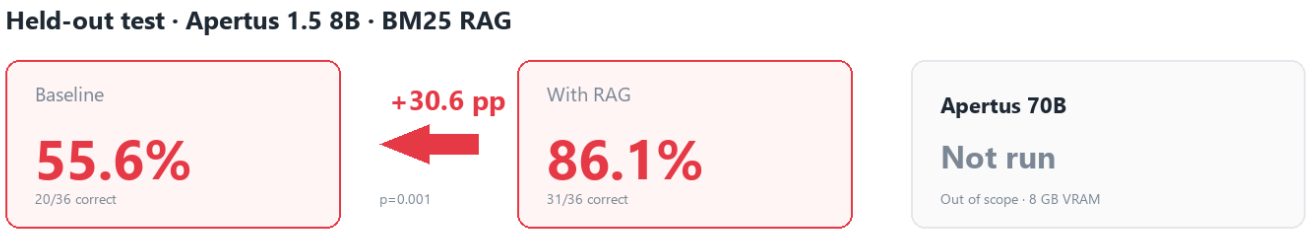


Figure C. Paired before/after on the sealed 36-item test set. 70B column is scope, not silence.

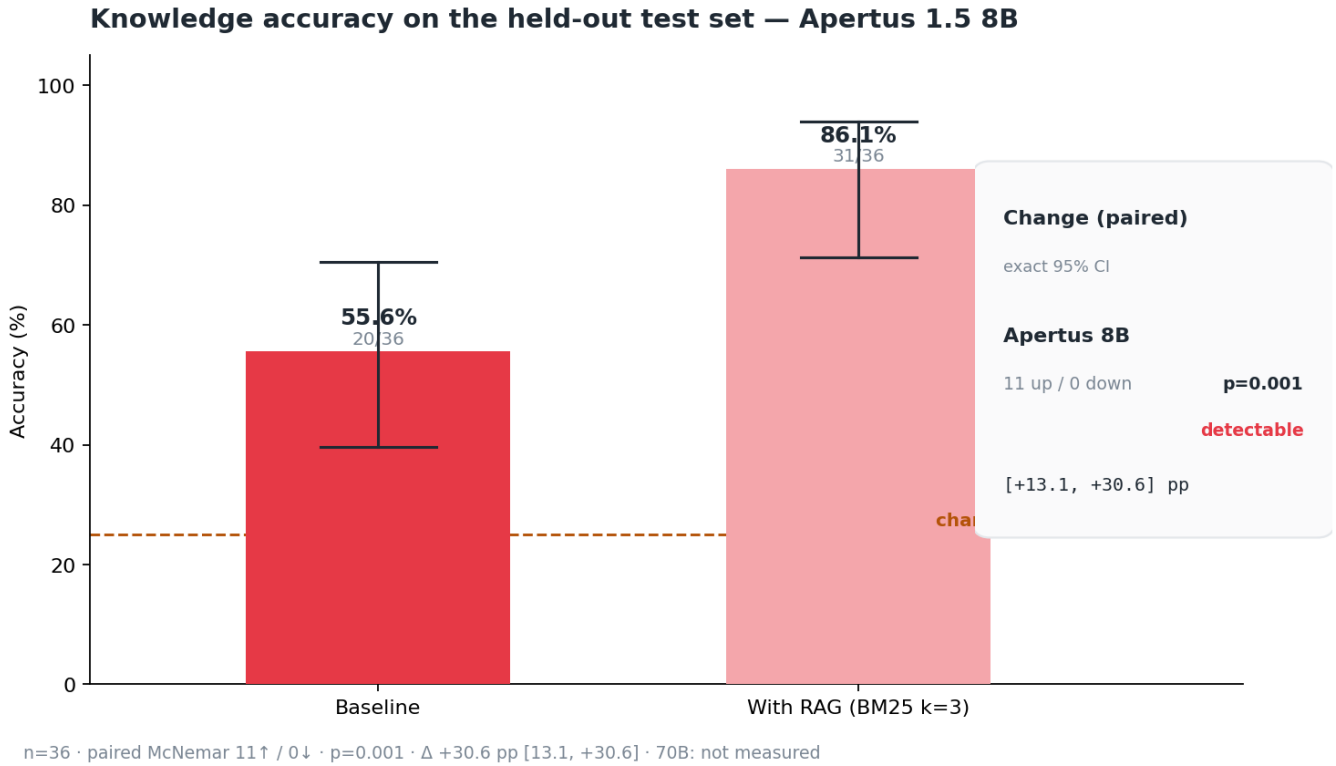


Figure 1. Overall accuracy with 95% Wilson intervals. Chance = 25%. Paired delta +30.6 pp, exact 95% CI [+13.1, +30.6]; 11 improved / 0 regressed; McNemar p=0.001.

Arm	Accuracy	95% Wilson CI	Notes
Baseline (no retrieval)	55.6% (20/36)	[39.6, 70.5]	8B measured
Improved (BM25 RAG k=3)	86.1% (31/36)	[71.3, 93.9]	8B measured
Paired delta	+30.6 pp	[+13.1, +30.6]	McNemar p=0.001
Apertus 70B (any arm)	—	—	Not run · 8 GB VRAM

At n=36 a paired change needs ≥6 improvements with 0 regressions (+16.7 pp) to be detectable. This run cleared that floor cleanly.

5. Where it helps · retrieval · transfer

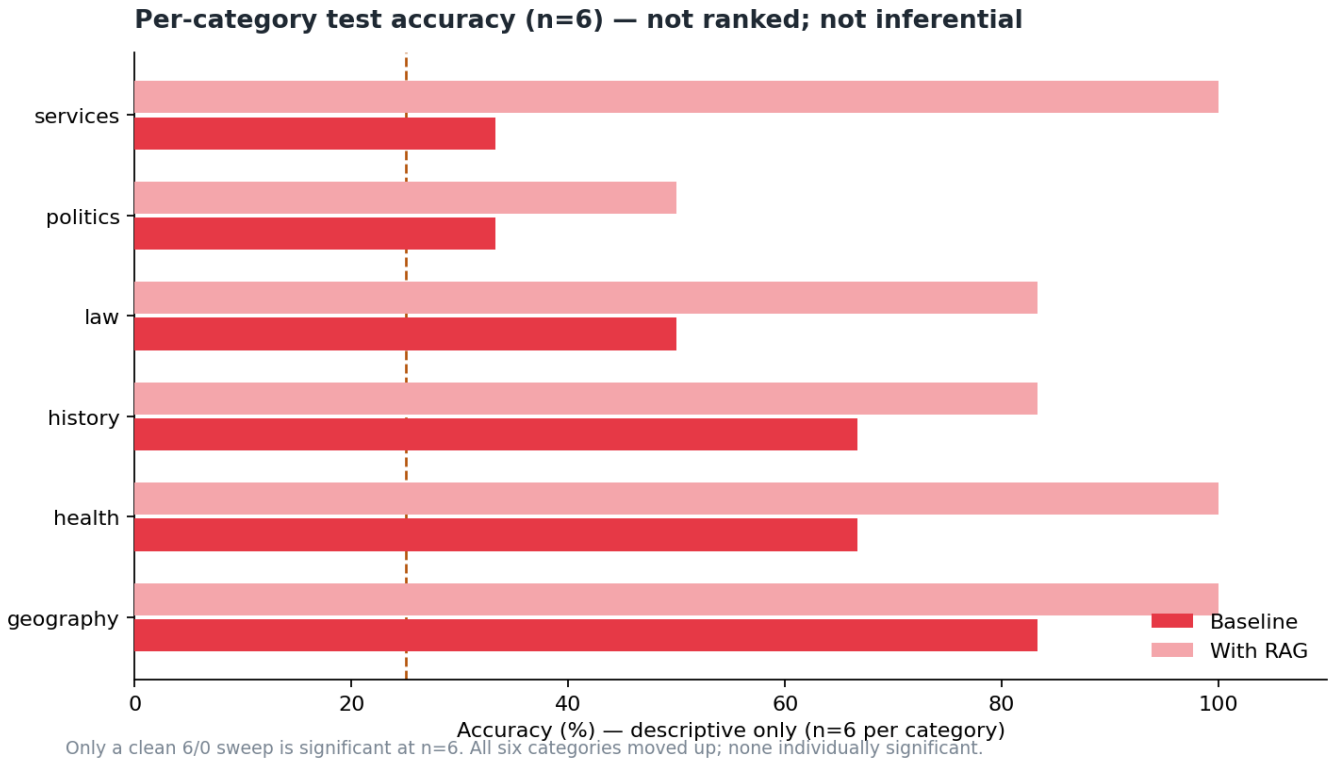


Figure 2. Per-category accuracies on the test set (n=6 each). Descriptive only — nothing except a clean 6/0 sweep is significant at this n. All six categories moved up; none is individually significant. Categories are alphabetical, not ranked.

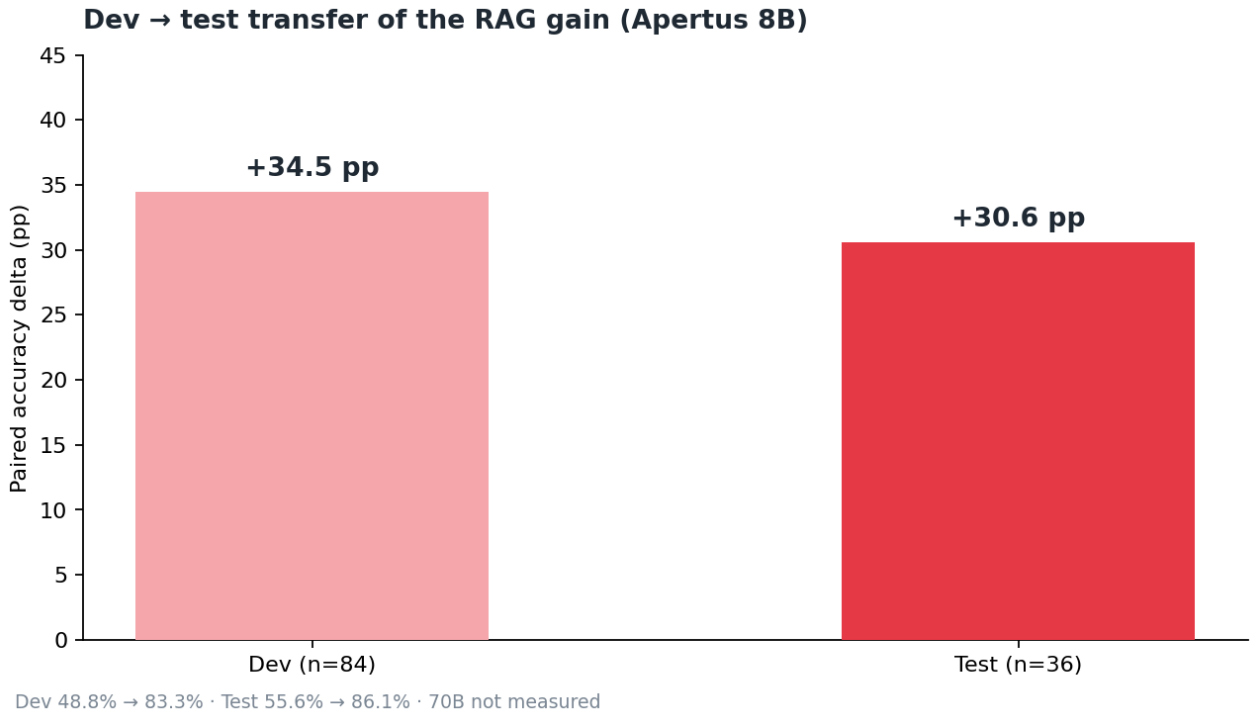


Figure 3. The RAG gain transfers from development (48.8% → 83.3%, +34.5 pp) to the sealed test (+30.6 pp). That is the scientifically interesting question the one-shot design was built to answer.

Retrieval recall on development (model-free): **83.3% / 90.5% / 94.0%** at k=1/3/5. Phase 4 used k=3. The contamination gate kept the behavioural system prompt at 0/84 answerable items vs 30% for a fact-list prompt.

6. Values, dialect, and the dual-model scoreboard

Evaluation scope — honest dual-model scoreboard

Metric	8B (measured)	70B
Knowledge baseline	**55.6% (20/36)**	Not run
Knowledge + RAG	**86.1% (31/36)**	Not run
Paired delta	**+30.6 pp · p=0.001**	—
Values mean / 2.0	**1.096 (self-judged)**	Not run
Dialect refusal	**50.0% (15/30)**	Not run

Figure D. Honest 8B | 70B scoreboard. Every 70B cell is out of scope — not imputed from 8B.

6.1 Values

73 judged records (30 English scenarios + multilingual renderings). Mean **1.096 / 2.0**; distribution {0: 33, 2: 40}; 0 unscored. The judge model is the same 8B weights under test (**self-judged**, disclosed per values_rubric.md §5). Human blind audit (B5, 15 items) was **not completed** — no agreement rate is claimed.

6.2 Dialect

30 items from Alemannic Wikipedia. On the live 8B test run: **50.0%** refusal (15/30); MCQ accuracy **50.0%** (18 items); translation mean **0.0** (all 12 translation items were refusals). All items carry `validated_by: null`. Refusal is reported as a headline metric, not folded into accuracy.

6.3 Why 70B is blank

Hardware: RTX-class 8 GB VRAM. Scope metadata records `dropped: ["70b"]` with reason matching this paragraph. The harness still accepts a 70B backend when available; this submission simply does not invent numbers for an arm that never ran.

7. Limitations

1. **n=36 is small.** Detection floor +16.7 pp with zero regressions. Per-category claims are not supportable at n=6.
2. **Single model scale.** No 8B-vs-70B comparison exists in the artefacts.
3. **Self-judged values.** No human agreement rate (B5 incomplete).
4. **Dialect unvalidated.** Native-speaker sign-off outstanding.
5. **Seal openings > 1.** Disclosed in eval/README.md; greedy decode, no between-run tuning; reported figures are from completed arms.
6. **Temporal.** Facts reflect Swiss law / institutions as of October 2026.

8. Reproducibility & distribution

- Git commit (stats): `436ba260830af773434675282a2cc897f15d13c9`
- Seed 42 · temperature 0.0 · prompt version 1.0
- Sealed test SHA-256: `df819e8c1582902a…`
- Code & results narrative: `https://github.com/bhuwanb23/apertus`
- Dataset (HF): `https://huggingface.co/datasets/bhuwanb23/apertus`
- License: MIT · Datasheet: DATASHEET.md (Gebru et al. 2021 format)
- Rebuild charts: `python scripts/prepare_submission_assets.py`
- Rebuild this PDF: `python scripts/build_submission_pdf.py`

Swiss Reality Check shows that a contamination-safe, source-grounded RAG stack moves Apertus 8B from barely above chance-plus-guesswork to high accuracy on verified Swiss knowledge — and that the gain survives a sealed hold-out. Scale-up to 70B remains an open measurement, not a claimed result.

Author: Bhuwan B · Hack Apertus Online Phase · 16 October 2026